rezki ::= engineer | educator

# Rezki Boureghdad

Back-end & Data Architect · Computer Science Educator

Algiers, Algeria
+213 542 708 547
boureghdadrezki@gmail.com
linkedin.com/in/boureghdad-rezki-309720268
github.com/V0idSeeker

## PROFILE

Software engineer focused on back-end and data architecture, and a computer-science educator. At Solrus, an Algiers software studio, I design the data models, APIs and integrations behind client production platforms. Before that I spent two years teaching 12 subjects at Université M'Hamed Bougara, from algorithms and databases to compilation and semantics, at Bachelor's and Master's level. I take the same approach in both roles: get the structure right, make it measurable, and explain it clearly.

## EXPERIENCE

### Data Specialist, Structure & Architecture — Present

Solrus — software studio for AI, cloud and secure systems · Algiers, remote-first

- Own back-end and data architecture for client platforms: domain modelling, relational schemas and migrations, API contracts, and third-party integrations.
- Build event-driven and async Python services (FastAPI, SQLAlchemy 2, Kafka, Redis, Celery) and tune them for latency and reliability.
- Security in the design from the start: role-based access, 2FA, audit trails, data-residency compliance, hardening before audits.
- Test-first delivery with pytest against real PostgreSQL and end-to-end Playwright suites. Work closely with front-end, AI and product teammates.

> Most client projects are under NDA. Names and details are not disclosed.

### Freelance Software Engineer (auto-entrepreneur) — Ongoing

Independent · web, mobile and AI products

- Run projects end to end for local businesses and community organisations: scoping, proposal, clickable prototype, build and deployment.
- Member of the team behind **Zakat.business**, an award-winning non-profit startup project.

### Teaching Assistant — Computer Science — 2022 – 2024

Université M'Hamed Bougara, Boumerdès

- Taught 12 subjects over two years to Software Engineering and LMD students, Bachelor's and Master's level.
- Prepared and ran tutorials and labs, wrote exercises and assessments, and helped students connect theory to hands-on practice.
- Adjusted my teaching for each audience, from first programming courses to formal semantics for Master's students.

### Trainer — Technical Workshops — University

DEVUP Club, Université M'Hamed Bougara

- Designed and ran practical software-engineering workshops for students, with hands-on material that participants applied to real projects.

## EDUCATION

**MSc Software Engineering & Data Science**
UMBB · Jan 2024

**BSc Computer Science**
UMBB · Jan 2022

## SUBJECTS TAUGHT

Algorithms & Data Structures
Databases
Programming Languages (intro → advanced)
Compilation
Semantics
+ 7 other CS modules

## CERTIFICATIONS

**NVIDIA**
Introduction to Machine Learning

**Neo4j**
Certified Professional

**EF SET English**
C2 Proficient

## LANGUAGES

**Arabic**
Native

**English**
C2 (EF SET)

**French**
Professional working

## SELECTED PROJECTS

Public, open and personal work only.

### Forsa / Aamal ATS — recruitment platform

Solrus public case study

- Connects companies, independent recruiters and job seekers in Algeria. Semantic job matching with sentence-transformer embeddings and location search with PostGIS.
- WebSocket chat, community forum, FCM push notifications, SlickPay subscriptions, JWT/Argon2 auth. Flutter mobile app and admin panel.

FastAPI · PostgreSQL/PostGIS · Flutter · Docker

### e-motion — back-office for cash-on-delivery shops

Own product

- Orders, stock, couriers and expenses, with real profit per order. Weighted-average purchase cost, stock that can never go below zero, alerts for customers who often cancel or return, and an audit log.
- Excel import, packing slips, English/French interface. Runs on free hosting tiers.

Next.js · TypeScript · Supabase · Vercel

### Ahl Quran — multi-tenant school manager

Community

- Study circles, attendance, student progress, messaging and reports, separated by organisation. Web and mobile apps.

FastAPI · PostgreSQL · Flutter/GetX · Docker

### 7alkaty — mosque course & imam-following app

Mobile

- REST API documented with Swagger, role-based access, scheduled push notifications, media storage, and a React Native client.

Node/Express · MongoDB · Expo

### Forest-fire detection & reporting

Master's thesis

- Co-designed a three-part system: a citizen reporting app, a data platform and alerting. Detection models built with YOLO and Faster R-CNN.

Python · YOLO · R-CNN · Flutter · Firebase

### ISO 27001 skill for Claude Code

Tooling

- Checks a codebase against ISO/IEC 27001:2022, plans the rollout step by step, and checks every edit with a scanner hook.

Python · Security compliance · AI tooling

## SKILLS

**Back-end**
Python, FastAPI, Django/DRF, SQLAlchemy 2 (async), Alembic, Pydantic, Celery, Node/Express

**Data**
PostgreSQL/PostGIS, Redis, Kafka, MongoDB, Neo4j, Elasticsearch, Supabase

**AI / ML**
RAG, LangGraph, FAISS/Chroma, embeddings & reranking, LLM APIs, YOLO/R-CNN

**Front & mobile**
TypeScript, Next.js, React, Tailwind, Flutter/Dart, React Native

**Ops & quality**
Docker, nginx, GitHub Actions, Vercel, Grafana/Loki, pytest, Playwright

**Security**
JWT, Argon2, OAuth2, TOTP 2FA, ISO 27001, data residency

**Also**
Java, C++, SQL; contract-first API design, decision records, AI-assisted development

## CS FOUNDATIONS

Algorithms & complexity

Compilers & formal languages

Programming-language semantics

Database theory & graph databases